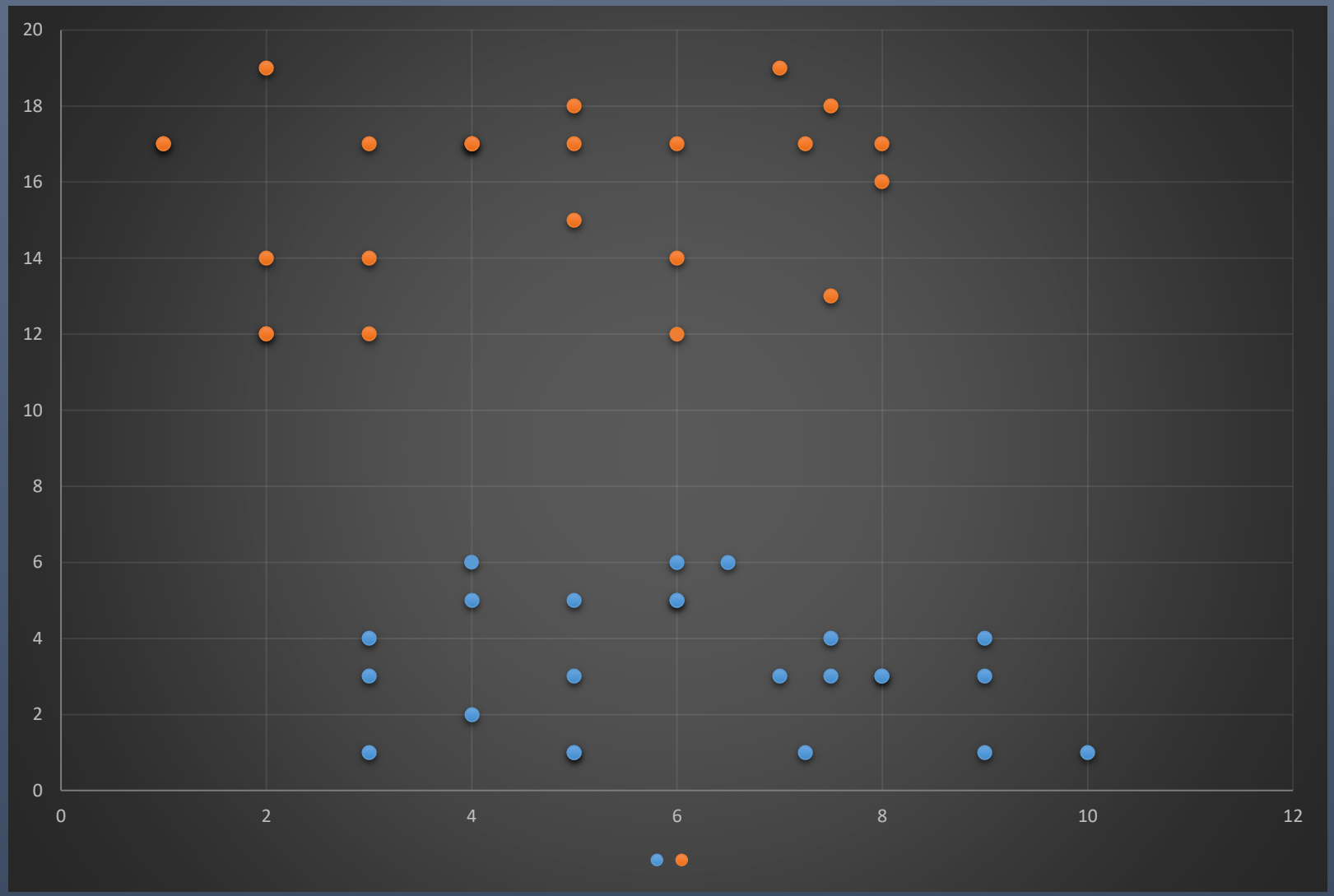
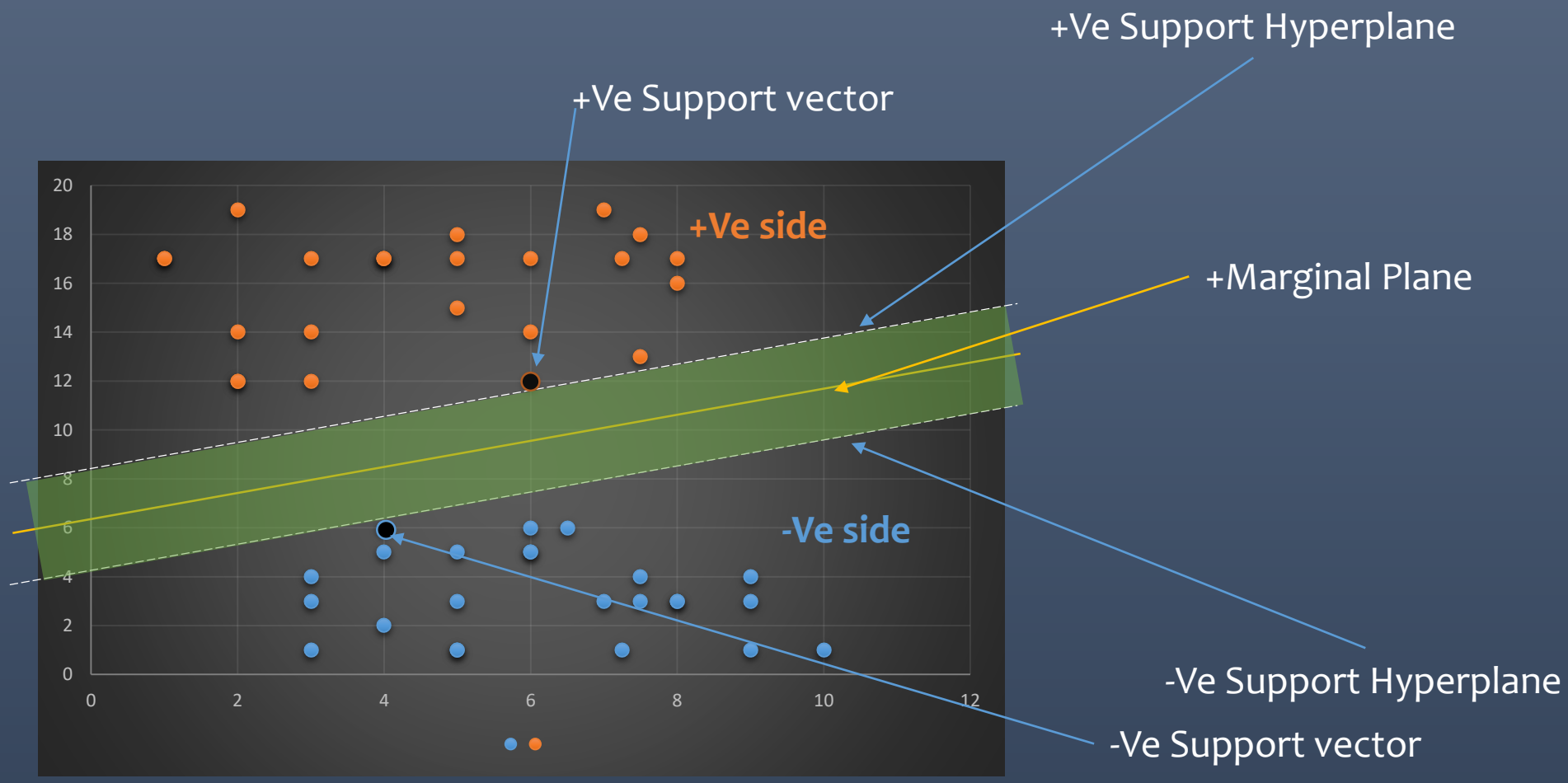

20
18
16
14
12
10
8
6
4
2
0
0
2
4
6
8
10
12

Support Vector Machine:

Support Vector Machine is used for Classification(SVC) as well as Regression(SVR) problems. However, primarily, it is used for Classification problems.

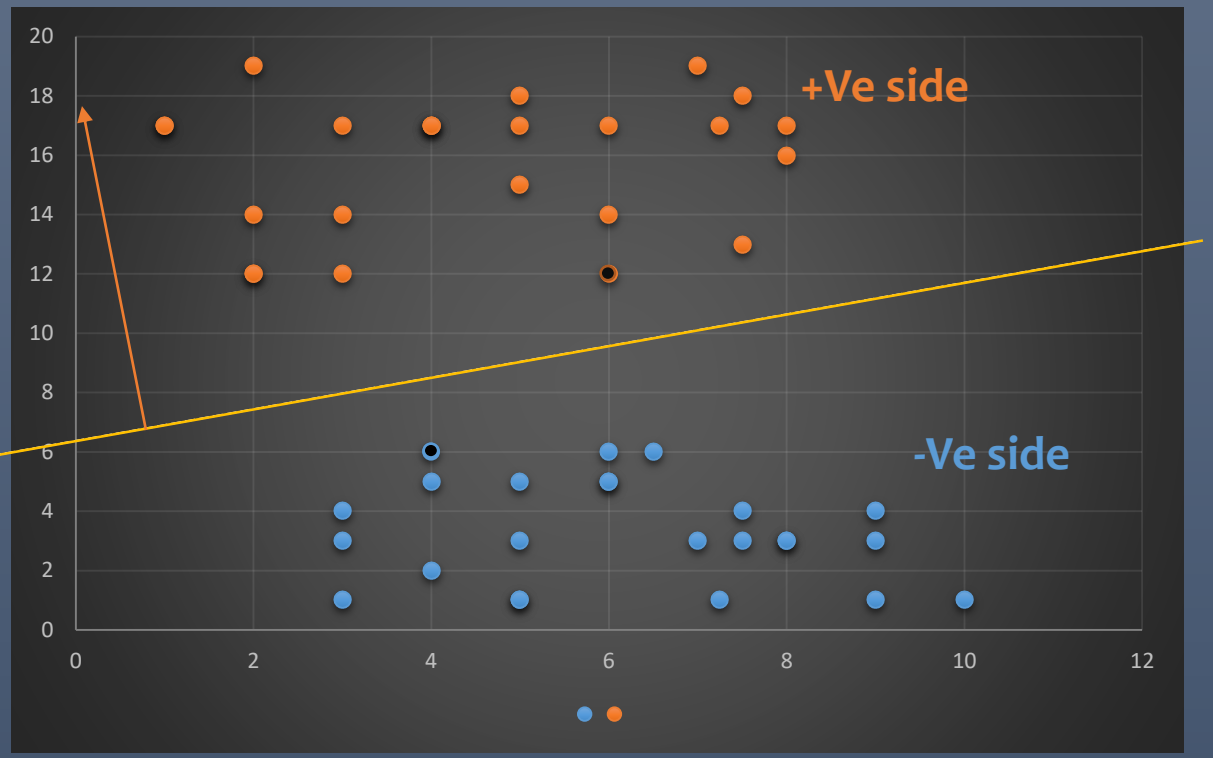


Support Vector Machine:

Support Vector Machine is used for Classification(SVC) as well as Regression(SVR) problems. However, primarily, it is used for Classification problems.

The primary objective of an SVM is to find the **optimal hyperplane** that best **divides the data points of different classes while maximizing the margin between the two classes**.

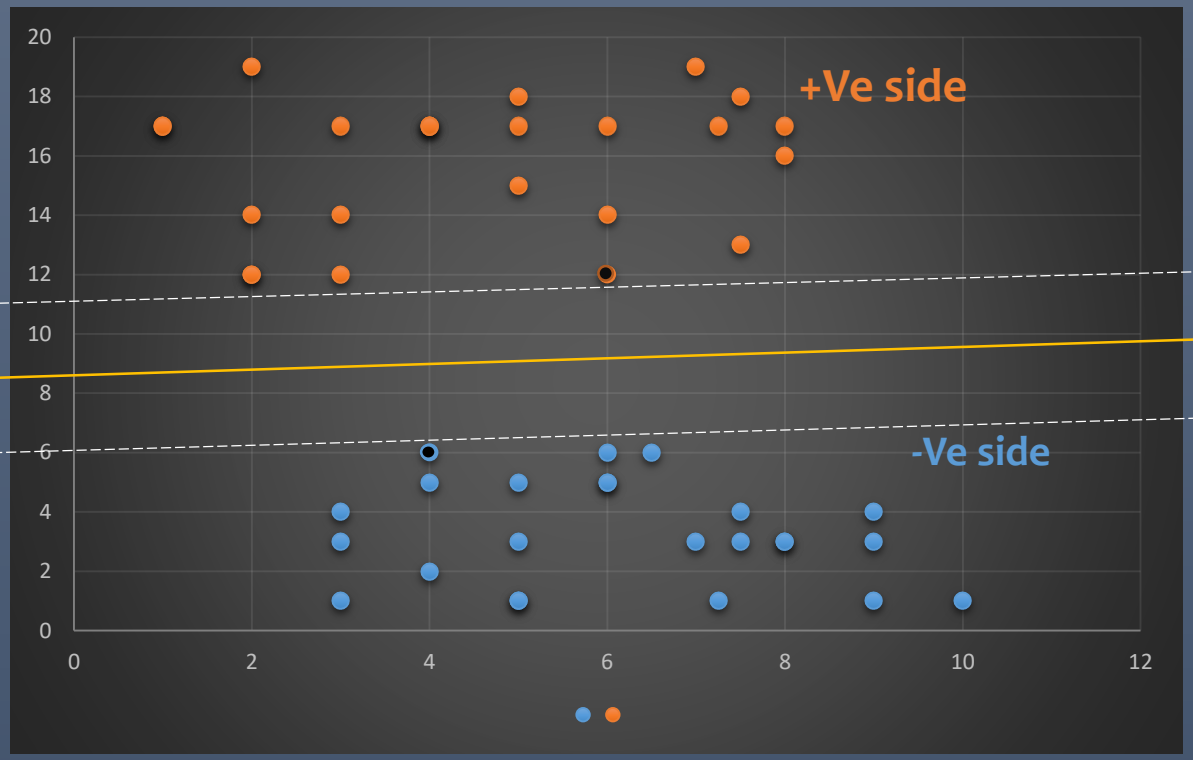
The margin is defined as the distance between the hyperplane and the nearest data points of each class (known as support vectors). SVM aims to find the hyperplane that not only separates the classes but also generalizes well, to new unseen data.



$$w^T x \quad + b \quad = 0$$

Geometrical intuition for SVC:

1) Create the hyperplane(line) which divides the points.
2) Find the points that falls on the +ve side and -ve side of the Hyperplane
3) Then find the +ve support vector (nearest point to the support Vector)
4) Followed by –ve Support vector (nearest point to the support Vector)
5) Basis of the + Support vector create +ve Hyperplane
6) Now create -ve Hyperplane.



$$w^T x_1 + b = 1$$

$$w^T x + b = 0$$

$$w^T x_2 + b = -1$$

$$w^T x_1 + b = 1$$

$$-$$

$$w^T x_2 + b = -1$$

$$w^T x_1 + b = 1$$

$$-(w^T x_2) + (-b) = -(-1)$$

$$w^T x_1 + b = 1$$

$$w^T(-x_2) + (-b) = 1$$

$$w^T(x_1 - x_2) = 2$$

We divide by ||w|| to **normalize** the weight vector into a unit vector, ensuring we are measuring the true perpendicular geometric distance across the margin rather than a slanted diagonal length

$$w^T(x_1 - x_2) = 2$$

$$\frac{w^T(x_1 - x_2)}{||w||} = \frac{2}{||w||}$$

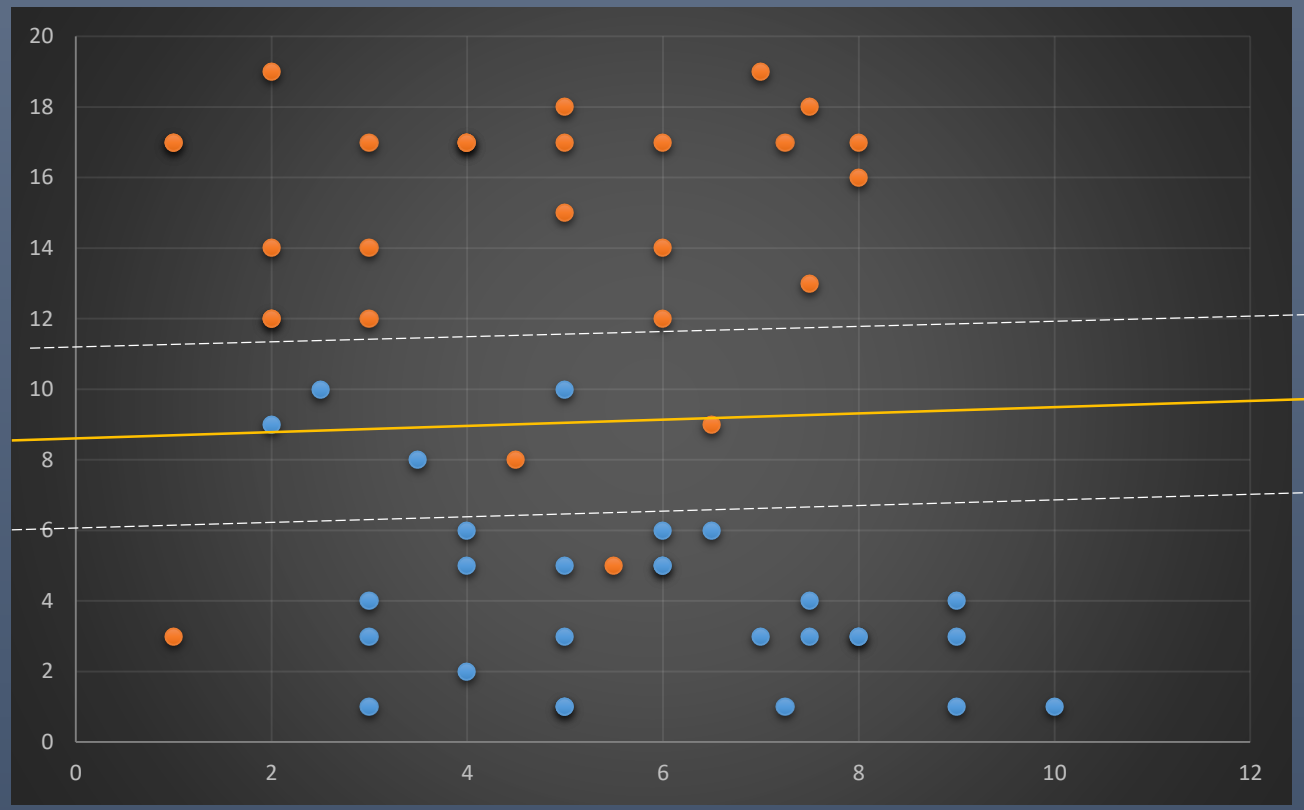
$$\text{Final equation} = \frac{2}{||w||}$$

$$(w, b) = \frac{||W||}{2} + C_i \sum_{i=1}^{w} \zeta_i$$

C_i: $Sum\ of\ all\ incorreclty\ classified\ points$

ζ_i: Sum of the distance of all incorrectly classified points. Here the distance will be taken from the marginal plane.

$$Hinge\ Loss = C_i \sum_{i=1}^{w} \zeta_i$$



Hinge Loss: In machine learning, hinge loss is a loss function used for training classifiers. The hinge loss is used for "maximum margin" in support vector classification,

Soft Margin

To handle data with errors in the marginal plane, adjustments should be made to the equation. This can involve reducing the marginal plane distance and incorporating hyper-parameter techniques,

1) reducing the marginal plane can be achieved by using $\frac{||W||}{2}$, Instead of $\frac{2}{||W||}$.
2) Here hyper parameter techniques, such as addition C_i and regularization techniques such as the addition of ζ_i can be employed to penalize to reduce excessive incorrect classifications.

In summary, to tackle data with errors in the marginal plane and create a more accurate model, modifications to the equation, such as incorporating regularization techniques, are essential. This enables the model to strike a balance between maximizing the margin and reducing errors, resulting in improved performance.

The updated equation is as below.

$$(w, b) = \frac{||W||}{2} + C_i \sum_{i=1}^{w} \zeta_i$$

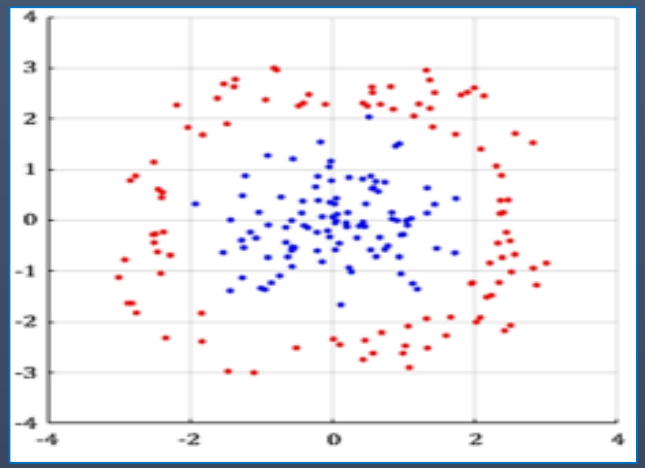
SVM Kernal?

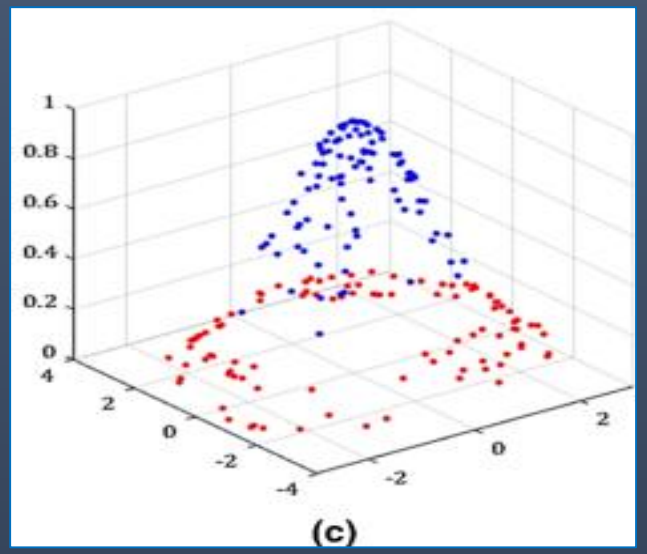
This allows SVM to effectively handle data that cannot be separated by a straight line.

In short, the SVM kernel is a mathematical function that allows SVM to handle complex, **non-linear data** by implicitly transforming it into a higher-dimensional space. Popular kernels include,

1) **Polynomial,**
2) **Radial basis function (RBF),**
3) **Sigmoid kernels,**

Each is suited for different types of data patterns. The choice of the kernel is crucial for SVM's performance, and it plays a key role in extending SVM's application to various classification problems.





$$(w, b) = \frac{||W||}{2} + C_i \sum_{i=1}^{w} \zeta_i$$

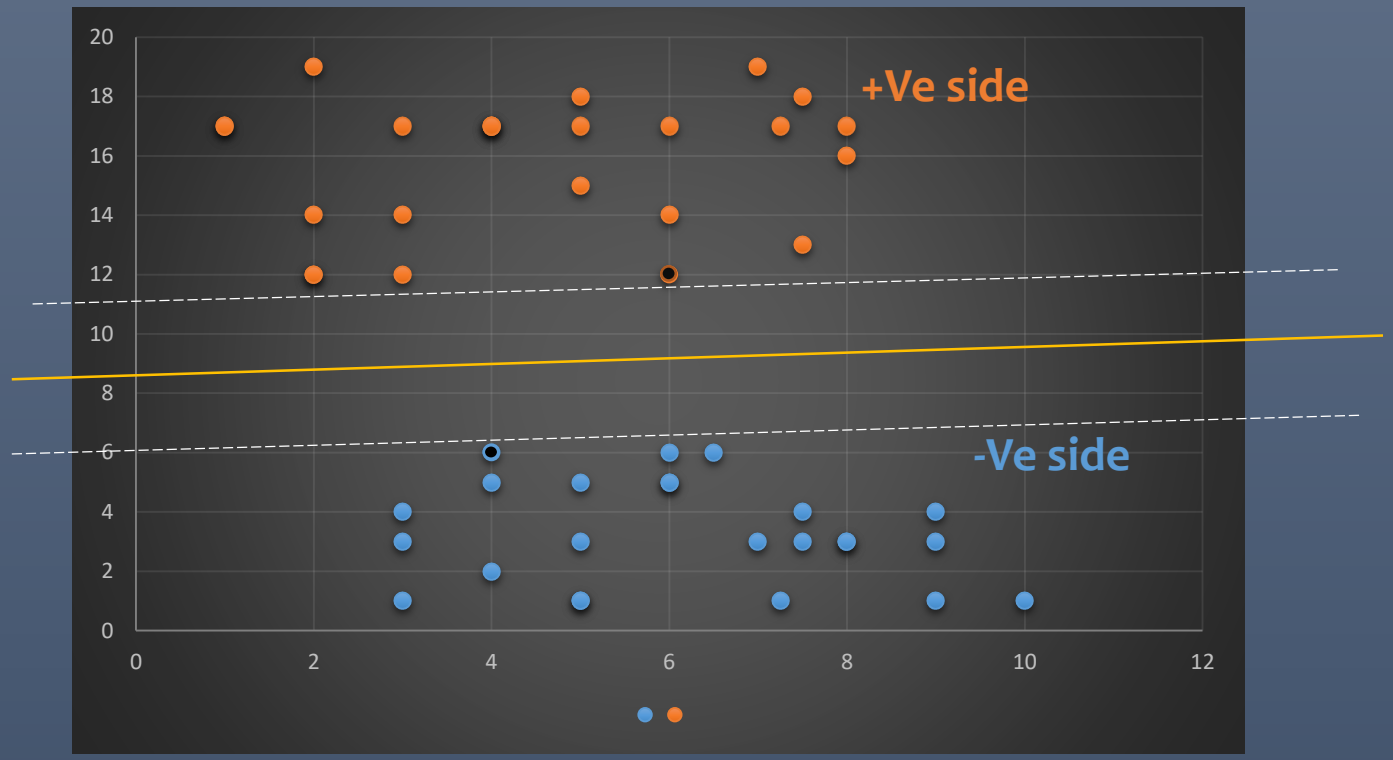
C_i: $Sum\ of\ all\ incorreclty\ classified\ points$

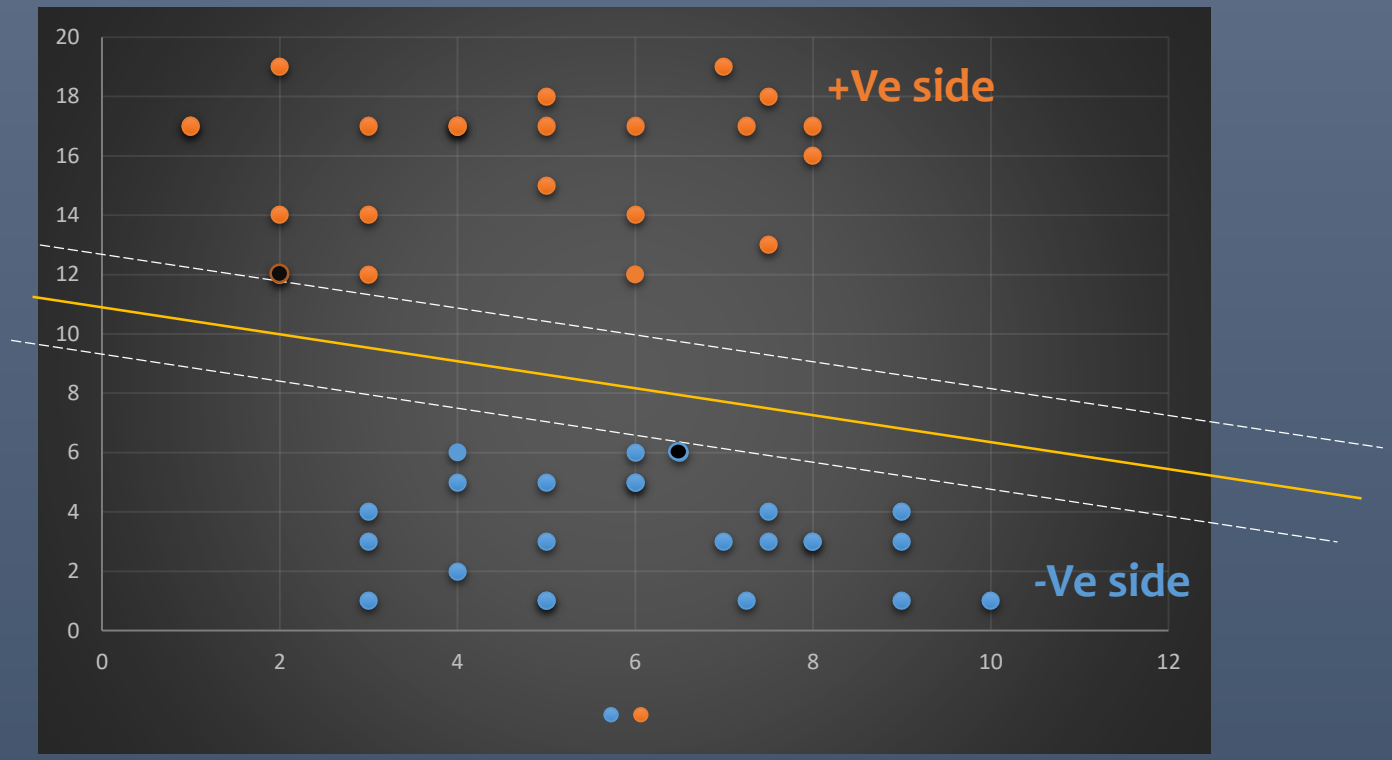
ζ_i: Sum of the distance of all incorrectly classified points.

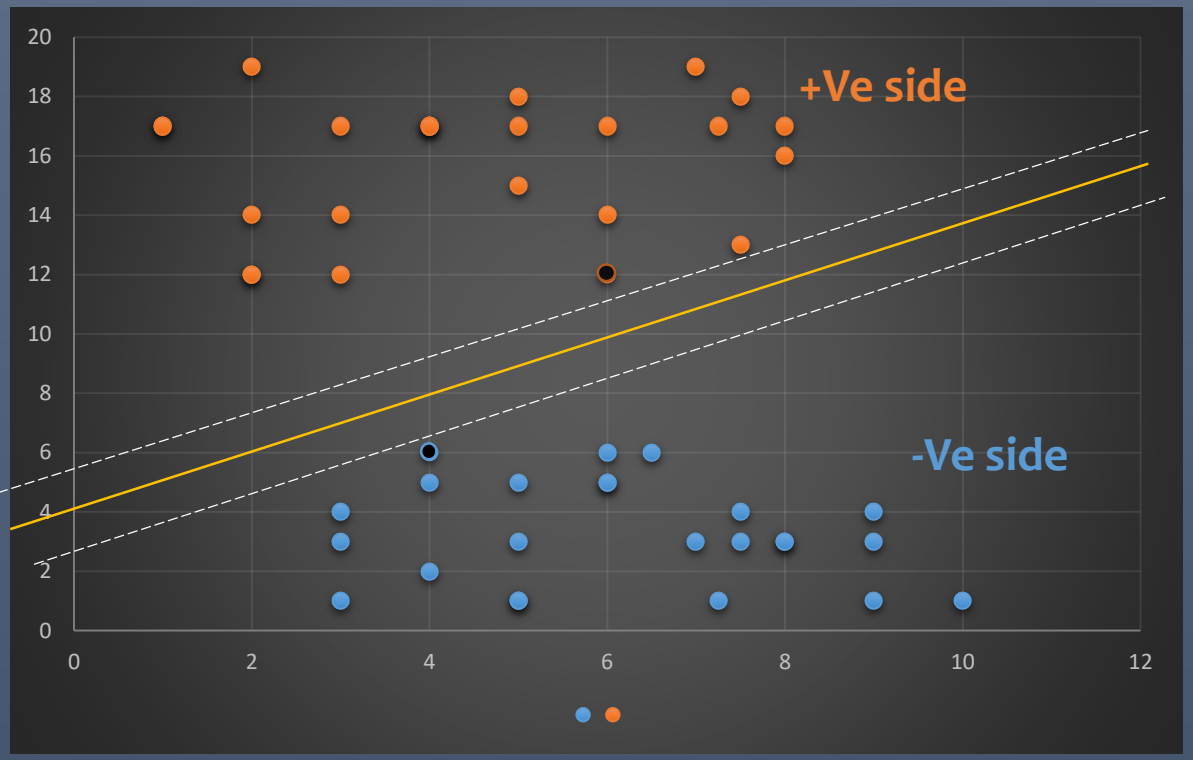
Here the distance will be taken from the marginal plane.

$$Hinge\ Loss = C_i \sum_{i=1}^{w} \zeta_i$$

Hinge Loss: In machine learning, hinge loss is a loss function used for training classifiers. The hinge loss is used for "maximum margin" in support vector classification,


+Ve side
-Ve side
20
18
16
14
12
10
8
6
4
2
0
0
2
4
6
8
10
12


+Ve side
-Ve side
20
18
16
14
12
10
8
6
4
2
0
0
2
4
6
8
10
12


+Ve side
-Ve side
20
18
16
14
12
10
8
6
4
2
0
0
2
4
6
8
10
12